

ITU-T

ICTs and Climate Change

Meeting with Mr. Monga, Director,
Energy and Climate Change Branch &
Mr. Leuenberger, Environmental
Management Branch
(Cleaner Production Centres), UNIDO

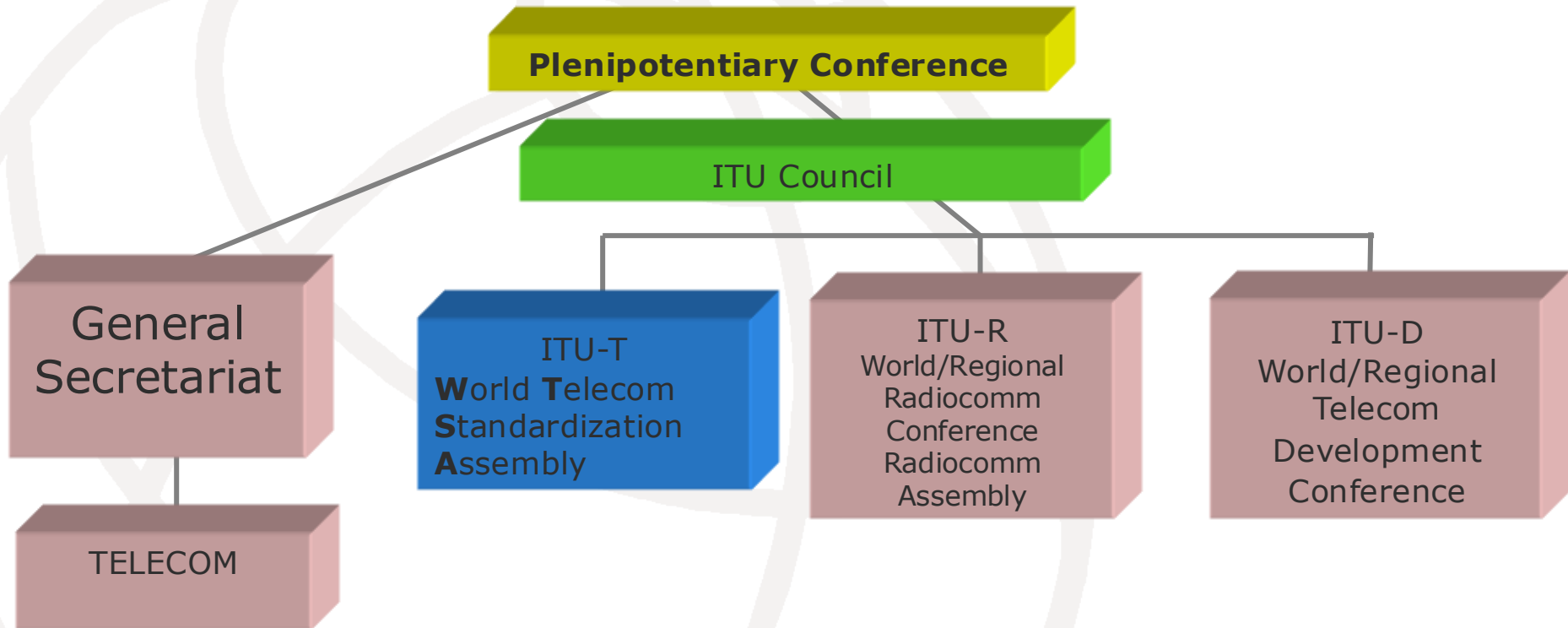
Introduction to ITU

- Founded in 1865, oldest specialized agency of the UN
- Standards making one of the ITU's first activities
- 191 Member States, 780 private sector entities
- HQ Geneva, 11 regional offices, 760 staff / 80 nationalities
- Named as one of the world's ten most enduring institutions by Booz Allen
- Five elected officials:
 - Secretary-General
 - Deputy Secretary-General
 - Director of the Radio Bureau (BR)
 - Director of the Telecommunication Standardization Bureau (TSB)
 - Director of the Telecommunication Development Bureau (BDT)

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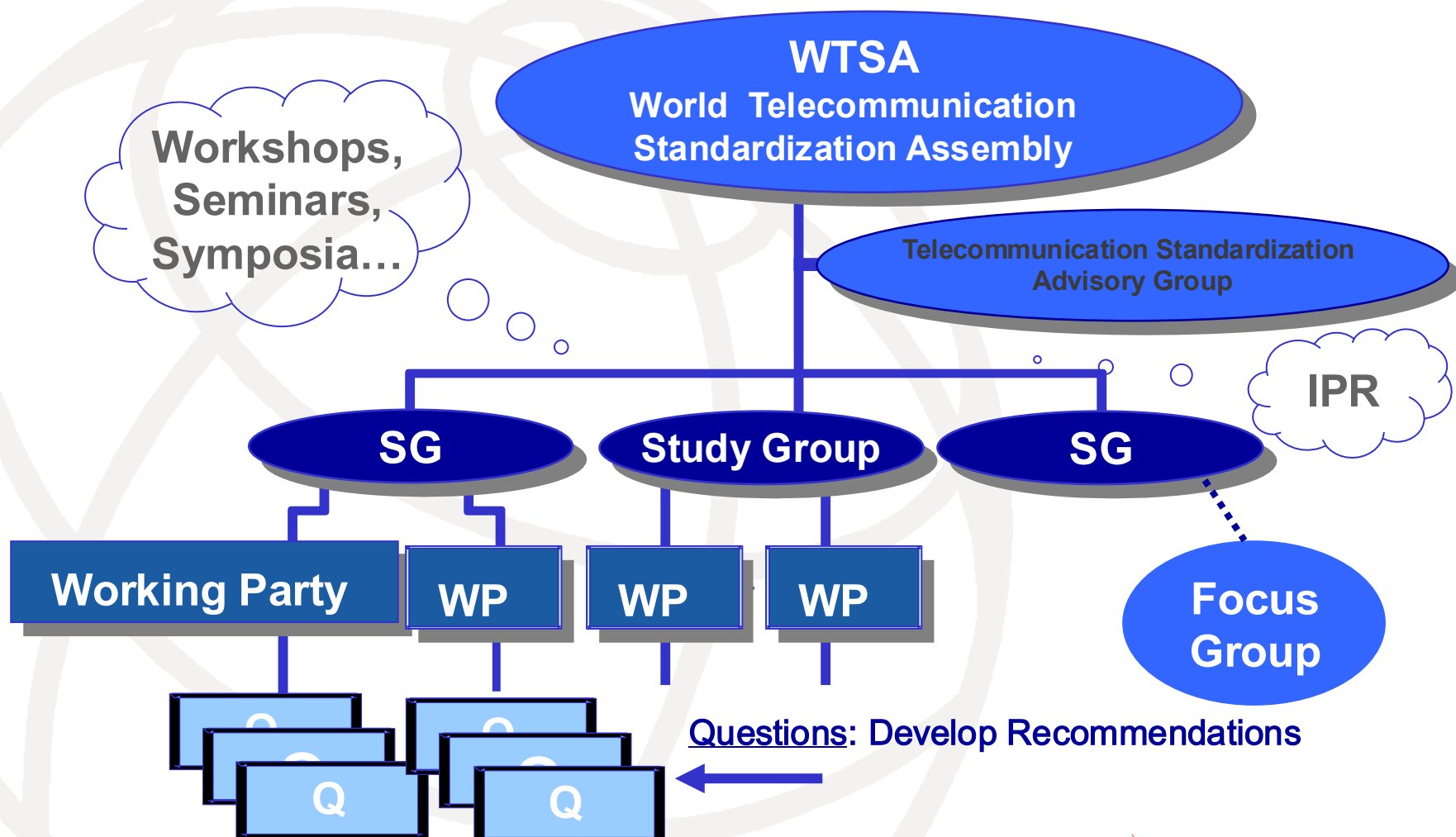


ITU Structure



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ITU-T Structure



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ITU-T Recommendations connect the world...

Without ITU-T standards you
couldn't make a telephone call
from one side of the world to
another.

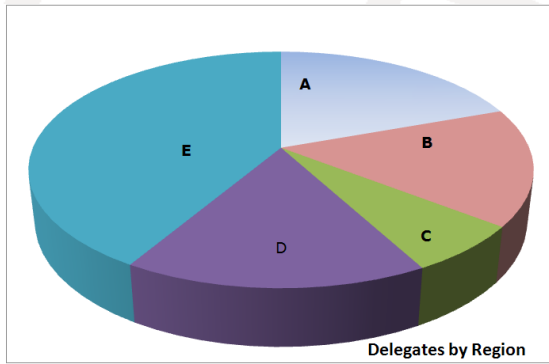
Without ITU-T standards the
Internet wouldn't function.



ITU-T Recommendations: Not all standards are equal

- Recommendations become mandatory if adopted in law
- Private standards may confuse users and consumers
- ITU's broad range of stakeholders, and robust processes provide the basis for consensus across sectors and countries
- Market-driven international standards, based on objective information and knowledge
- Meet the needs and concerns of all relevant stakeholders

Member State Participation



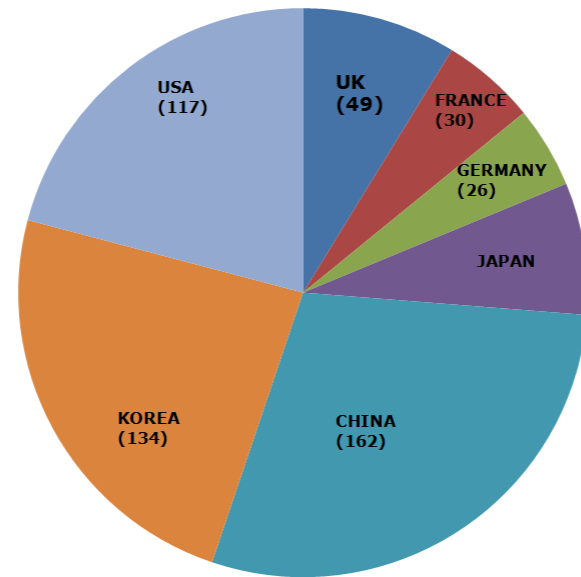
Region A - The Americas (216)

Region B - Western Europe (178)

Region C - Eastern Europe and Northern Asia (73)

Region D - Africa (182)

Region E - Asia and Australasia (460)



Strategic Objectives

1. Develop and publish timely global standards
2. Identify relevant areas for future standardization projects
3. Provide the most attractive forum for standardization in the interest of the membership
4. Promote value of ITU-T to attract increased membership
5. Disseminate information and know-how
6. Cooperate and collaborate with other Sectors and other entities
7. Provide support and assistance to the membership, in particular developing countries

ITU-T Key Features

- Open, transparent, consensus based, fast working, public/private partnership
- Technical standards developed by industry members, when consensus placed on website and if no comments after 4 weeks is in effect approved by 191 governments
- ITU standards are therefore truly global, open standards, available free of charge, unlike those of many other standards bodies, fora or consortium that claim to produce global and open standards. Publicly available database of products and services meeting ITU standards
- Organizing interoperability events to prove interoperability of different vendors equipment
- Conform
- Common IPR policy with ISO and IEC (FRAN)



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Importance of Global Standards

- Global Standards essential in a complex world
- Standards make things easier
- Essential for international communications and global trade
- Drive competitiveness, for individual businesses and world economy
- Help organizations with their efficiency, effectiveness, responsiveness and innovation
- Lower prices and increase availability by reducing technical barriers and promoting compatibility between systems and networks
- Manufacturers, network operators and consumers benefit

Standards proven economic tool

- WTO trade report 2005
- British Standards Institute (BSI): standards make annual contribution GBP 2.5 billion.
- German standards body (DIN): economic benefits standardization about 1% GDP.
- Canada: 17 % of labour productivity increase and nine per cent of growth of GDP 1981-2004.
- Standards have a significant effect on limiting the undesirable outcomes of market failure.
- The work of ITU has smoothed the more economical introduction of new technologies.

ITU-T collaboration

44 formal partnerships



The Internet Corporation for Assigned Names and Numbers



- Vienna Agreement between the international standards orgs and their European regional counterparts.
- World Standards Cooperation
 - Patent policy & Joint events
- ITU-T and IEEE
 - MoU & Joint events
- Global Standards Collaboration
 - Supports ITU as preeminent global ICT standards organization.
- ITU-T and 3GPP
- ETSI
 - Management meetings
- ITU-T and IETF
 - Management meetings
- ITU-T and ICANN
 - Management meetings
- E-Business MoU: IEC, ISO, ITU and UN/ECE

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ITU and Climate Change



"Climate Change is a global challenge that the world cannot lose".

Dr Hamadoun I. Touré
ITU Secretary-General, 13 November 2008



"Climate change is the defining challenge of our era. ITU's work to cut greenhouse gas emissions, develop standards and use 'e-environment' systems can speed up the global shift to a low-carbon economy".

Ban Ki-moon
United Nations Secretary-General, 12 November 2008

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**UN Secretary-General,
Ban Ki-moon: "ITU is one of the very
important stakeholders in the area of
climate change."**

- Smarter standards for greener systems & services
- Focus Group of 20 organisations developed basis for methodology to estimate GHG emissions from ICTs Sector over their entire life-cycle



ITU and COP

- Ongoing role in UNFCCC process
 - Promotion of role of ICTs in reducing GHGs
- Side events and press conference in Barcelona and Copenhagen
 - Partners: WWF, WIPO, WMO, GeSI, OECD, UNFCCC, Analysys Mason, Cisco, Microsoft, NTT, Government representatives from Japan, Ecuador
- ISeeT Kiosk – Daily briefings from high level ICT business people
- High level meetings

Standardization (1)

- Developing a methodology to measure impact of ICTs on climate change (see later slide)
- Identifying priority sectors where ICTs can reduce emissions (e.g. smart buildings, intelligent transport systems)
- The promotion of NGNs (reducing power consumption by up to 40%)
- Lifecycle analysis and disposal/recycling of ICTs
- All new standards are now checked for energy efficiency

Standardization (2)

Standardization work
in the field of
Intelligent Transport
Systems, Smart
Grid, e-waste,
sensor-based
networks based on
RFID & telemetry



Universal charger

Delivers 50% reduction in standby energy consumption, eliminates 51,000 tonnes of redundant chargers, and cuts GHG emissions by 13.6 million tonnes annually



Methodology

- Common methodology for measuring ICT carbon footprint of ICT Sector ~ 3%
- Methodologies to estimate impact of ICTs on reducing emissions in other industry sectors ~ 15%
- Without, it will be impossible to provide meaningful comparisons.
- Helps to establish the business case to go green.

Reports, symposia, working methods

- TECHWATCH Reports on CC and positive impact of new technologies
 - Next Generation Networks, Intelligent Transport Systems, etc.
- Major Symposia on ICTs and CC
 - 2008: Kyoto and London
 - 2009: Quito and Seoul (virtual event)
- ITU-T pioneering energy efficient work methods
 - Paperless meetings, on-line work tools, virtual symposia.
- ITU-T leading Dynamic Coalition on Internet and Climate Change as part of the Internet Governance Forum

Radiocommunication

- **Monitor climate change by:**
 - Provides spectrum and orbital resources for satellite remote-sensing
 - Providing key climate data via radio-based applications (e.g. RFIDs)
 - Work closely with WMO

Development

- E-environment Toolkit for policymakers gives principles and guidelines for the development of applications and services in the area of the environment.
- Will help countries to assess the contribution that ICTs can make to reduce GHG emissions

ITU Resolutions on Climate Change

- WTSA-08, Resolution 73, resolves that CC is a high priority in ITU
- WTPF (April 2009), Opinion 3, instructs promotion of Res. 73.
- GSC-14 (July 2009), Resolution, encourages related collaboration, etc.
- ITU Council (Oct. 2009), Resolution 1307, unanimously decided its importance and active participation in UNFCCC including COP-15 in Copenhagen



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